

3(a)(i)	0.10 s or 0.30 s or 0.50 s or 0.70 s or 0.90 s	A1
3(a)(ii)	0 or 0.40 s or 0.80 s	A1
3(b)(i)	$\omega = 2\pi / T$	C1
	$= 2\pi / 0.40$ $= 16 \text{ rad s}^{-1}$	A1
3(b)(ii)	$v_0 = \omega x_0$	C1
	$= 15.7 \times 2.5 \times 10^{-2}$ $= 0.39 \text{ m s}^{-1}$	A1
	or	
	tangent drawn at steepest part and working to show attempted calculation of gradient	(C1)
	leading to $v_0 = 0.39 \text{ m s}^{-1}$ (<i>allow</i> $\pm 0.15 \text{ m s}^{-1}$)	(A1)
3(b)(iii)	$a_0 = \omega^2 x_0$	C1
	$a_0 = (15.7^2 \times 2.5 \times 10^{-2})$ $= 6.2 \text{ m s}^{-2}$	A1
	or	
	$a_0 = \omega v_0$	(C1)
	$a_0 = 15.7 \times 0.39$ $= 6.2 \text{ m s}^{-2}$	(A1)

Question	Answer	Marks
3(c)	period is shorter/lower	B1
	Any one from: • greater spring constant/stiffness • (restoring) force is greater (for any given extension) • acceleration is greater (for any given extension) • greater energy/maximum speed (for a given amplitude)	B1

Question	Answer	Marks
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